Forecasting Mainland China Inflation

Forecast accuracy, operational reliability, and Chinese New Year inference

DS607 Statistics for Data Science
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Individual course project

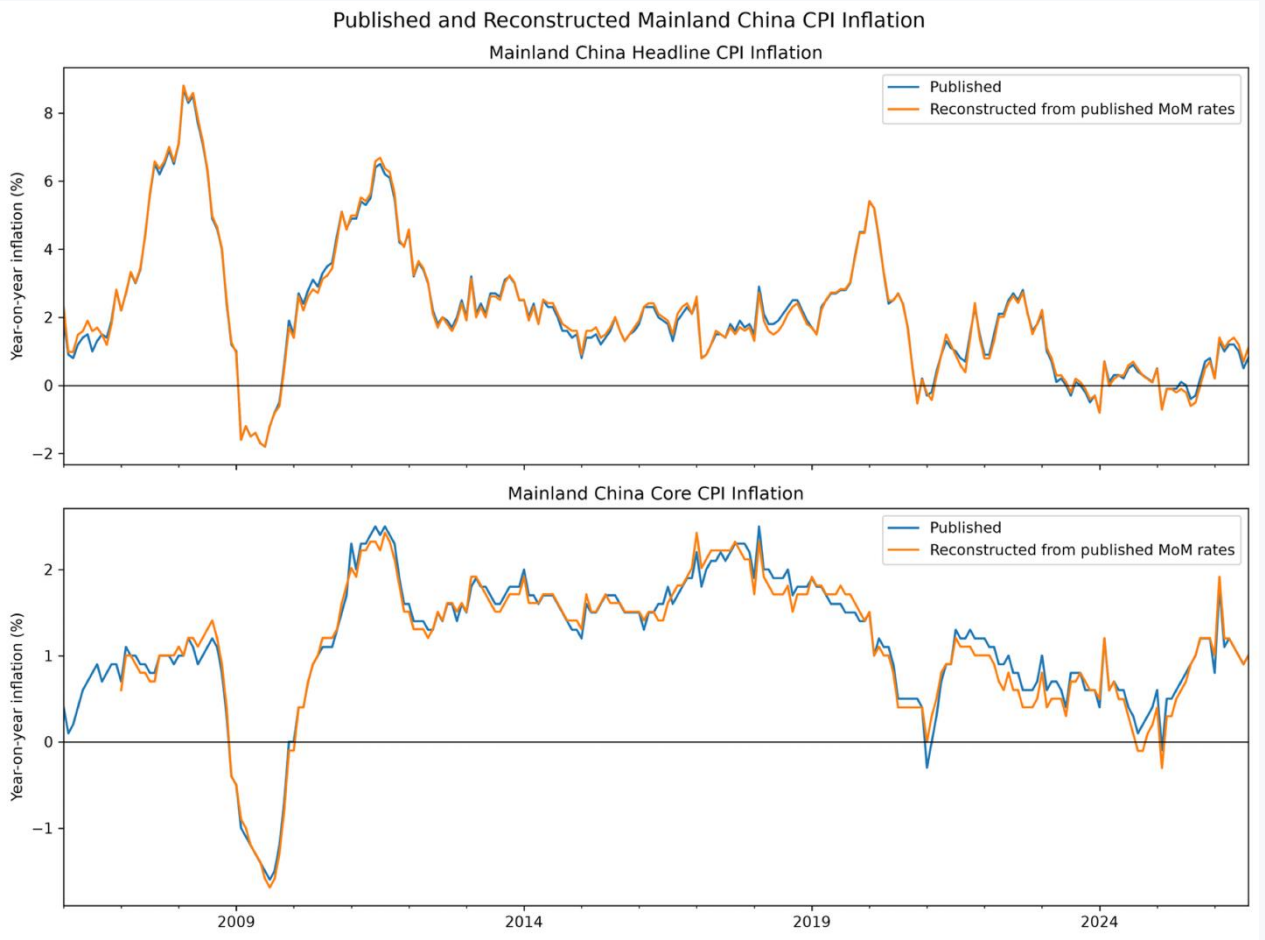
Forecast monthly inflation, evaluate year-on-year outcomes

NBS statistics accessed through CEIC; common headline/core sample starts January 2006

1. Model published MoM inflation
2. Compound into a CPI index path
3. Convert to YoY forecasts
4. **Compare with published YoY**

Why this matters

Models forecast monthly rates, but the decision-facing comparison is year-on-year inflation.



The reconstructed series track the published series closely enough for a coherent forecast transformation.

A locked, expanding-window evaluation avoids look-ahead bias

Development decisions end in 2014; final evaluation covers 117 attempted forecast origins

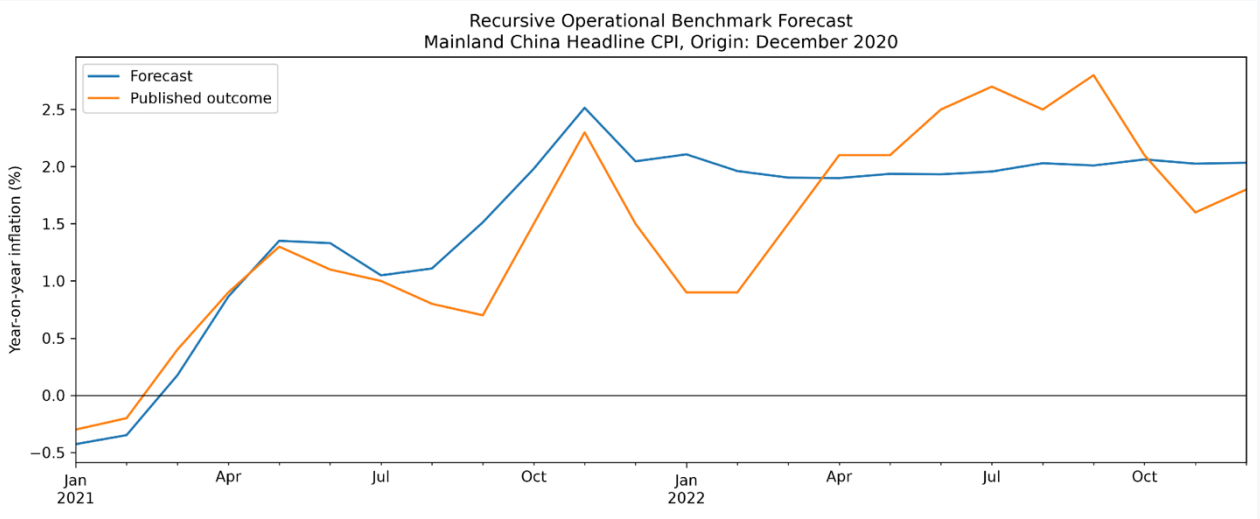
Development

Jan 2006 - Dec 2014
Select and lock models

Final evaluation

Origins: Dec 2014 - Aug 2024
Targets through Aug 2026

24	117	104
max horizon (months)	attempted origins	common successful origins



The benchmark uses the previous five same-calendar-month values and becomes recursive beyond 12 months.

The final comparison is deliberately narrow and interpretable

All specifications were locked before final-evaluation outcomes were examined

HEADLINE

Benchmark	5-year recursive same-month average
Conventional	SARIMA $(1,0,1)(1,0,1)_{12}$
CNY-aware primary	Seasonal-MA SARIMAX + month + CNY
CNY-aware robust	Seasonal-AR SARIMAX + month + CNY

CORE

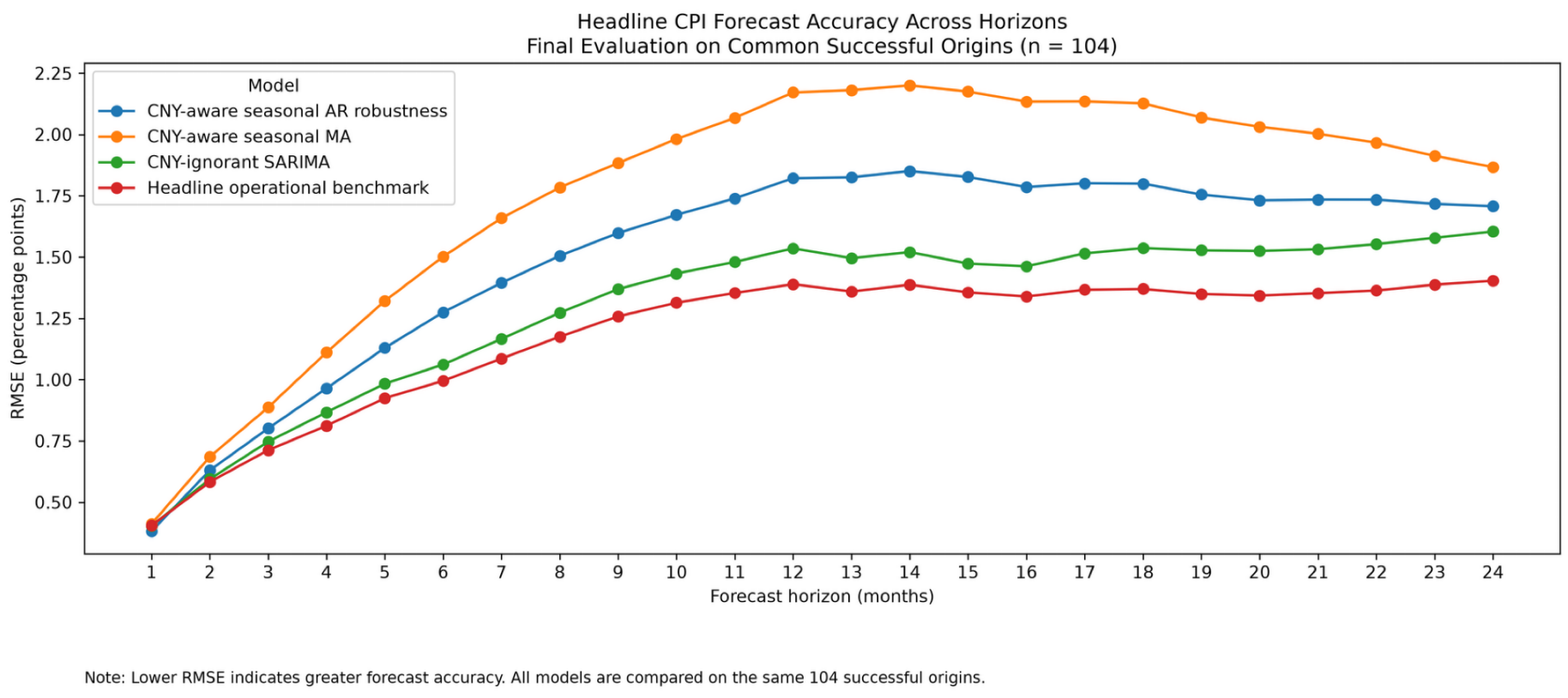
Benchmark	5-year recursive same-month average
Primary statistical	ARMA(1,1) seasonal ARMA
Reliability robust	MA(2) seasonal ARMA

Selection considered fit, residual diagnostics, valid roots, repeated-origin reliability, and development-stage forecast performance.

The project compares transparent low-complexity models, not a large machine-learning bake-off.

Headline: the benchmark wins beyond one month

Lower RMSE indicates better year-on-year forecast accuracy; n = 104 common successful origins



H1

CNY-aware seasonal AR has lowest RMSE

H2-24

Operational benchmark has lowest RMSE

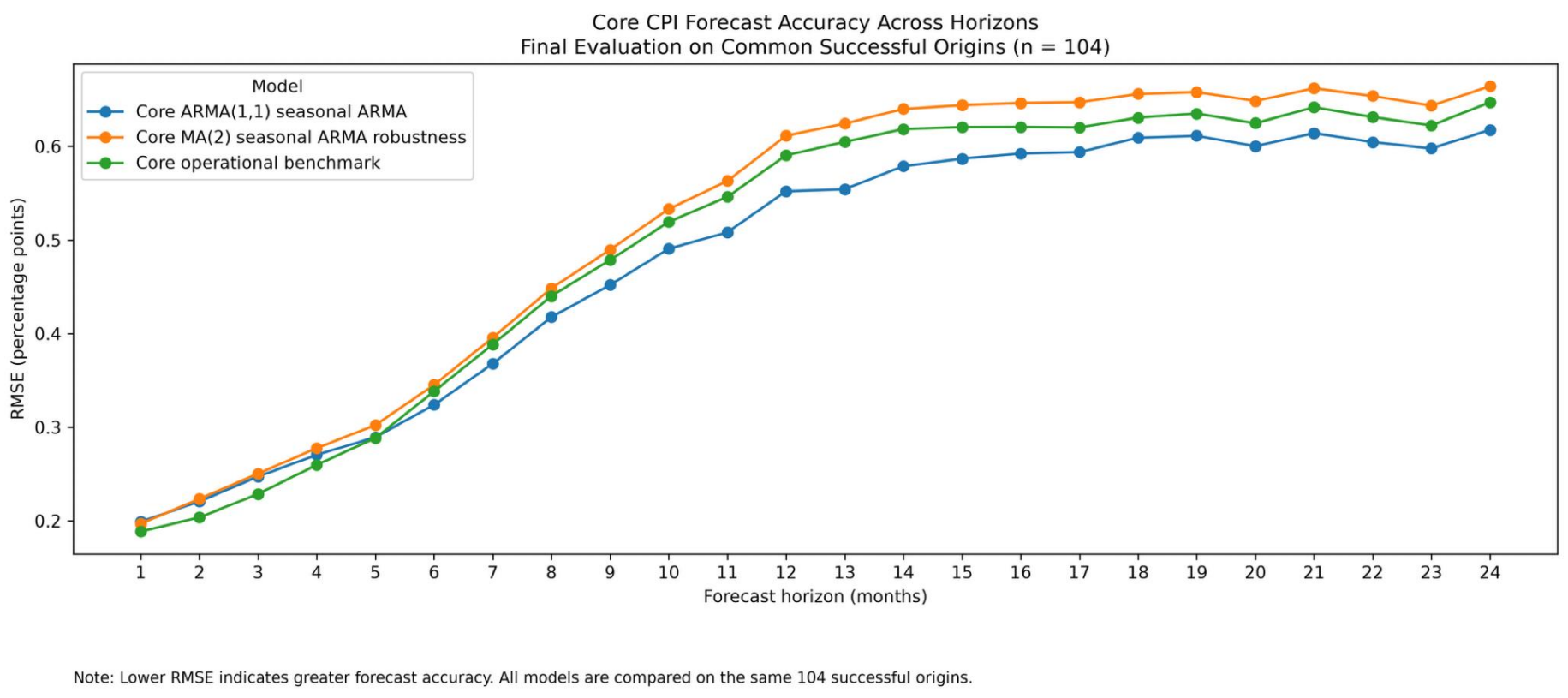
Bias grows with horizon

Negative mean errors indicate systematic overforecasting.

Adding CNY structure improves explanation, but does not deliver general headline forecast superiority.

Core: statistical modelling adds value from six months onward

The benchmark leads through H5; the locked ARMA(1,1) seasonal ARMA leads from H6 to H24



H1-5

Operational benchmark has lowest RMSE

H6-24

Core ARMA(1,1) has lowest RMSE

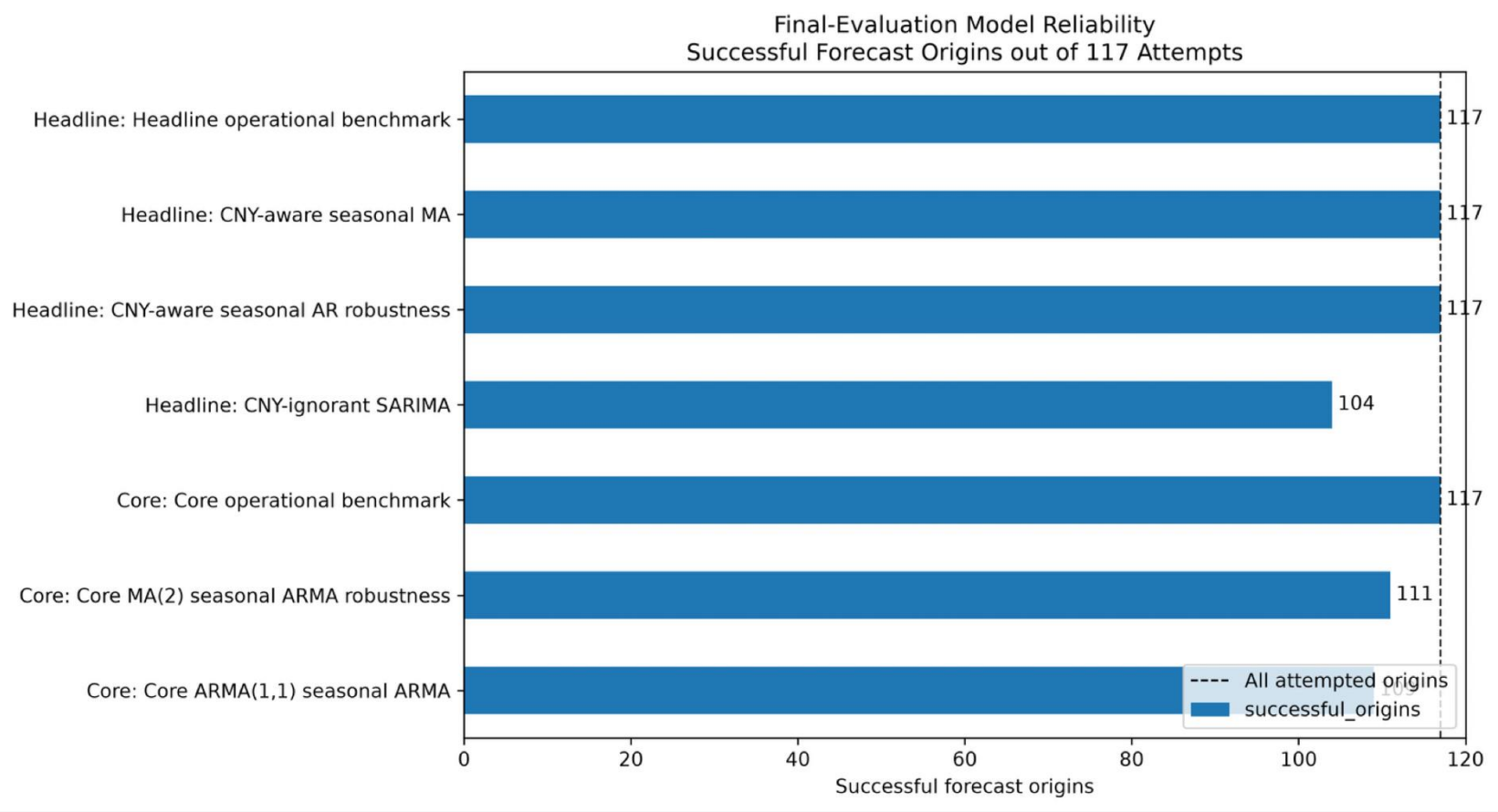
H24 RMSE

0.618 statistical vs 0.647 benchmark

The practical answer differs by target: simple for headline, statistical at longer core horizons.

Accuracy is conditional on models producing valid forecasts

Reliability is reported across all 117 attempted origins, separately from paired accuracy



100%
benchmarks

100%
CNY-aware headline

88.9%
CNY-ignorant headline

93.2%
core ARMA(1,1)

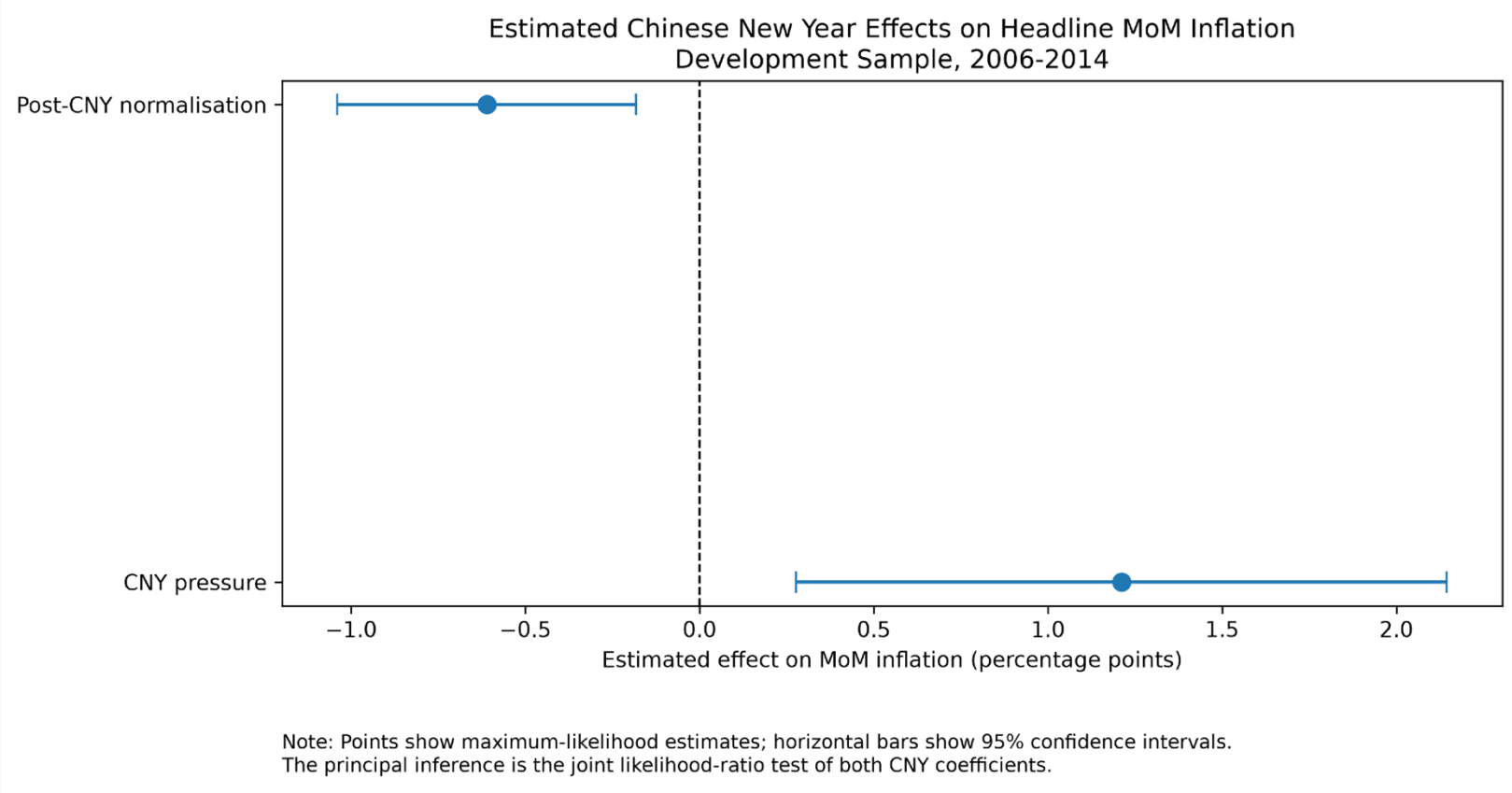
Why separate the metrics?

A low RMSE on successful origins does not compensate for frequent fitting failure.

Operational usefulness depends on both loss and the probability of obtaining a valid forecast.

CNY effects are explanatory, not a guarantee of forecast gains

Matched likelihood-ratio test on the 2006-2014 development sample



Joint LR test

LR = 28.985
df = 2
$p \approx 5.08 \times 10^{-7}$

Estimated effects

Pressure: +1.211 pp
Normalisation: -0.611 pp

Strong in-sample likelihood evidence coexists with weak medium- and long-horizon headline forecast performance.

Decision takeaway: use different tools for different purposes

Forecast accuracy, reliability, and explanatory inference answer distinct questions

HEADLINE

Default to the benchmark

Best RMSE from H2-H24
Perfect operational reliability

CORE

Use the statistical model at longer horizons

Benchmark through H5
ARMA(1,1) from H6-H24

CNY INFERENCE

Use CNY variables for explanation

Strong LR evidence
Not a forecast guarantee

Key limitations

Published rather than real-time vintages • reconstructed index levels • rounded MoM rates • deterministic CNY windows • some near-boundary roots and fitting failures

More structure can improve interpretation without improving operational forecasts.

Backup slides

For technical questions on specifications, CNY construction, and development-stage selection

Backup: exact locked specifications

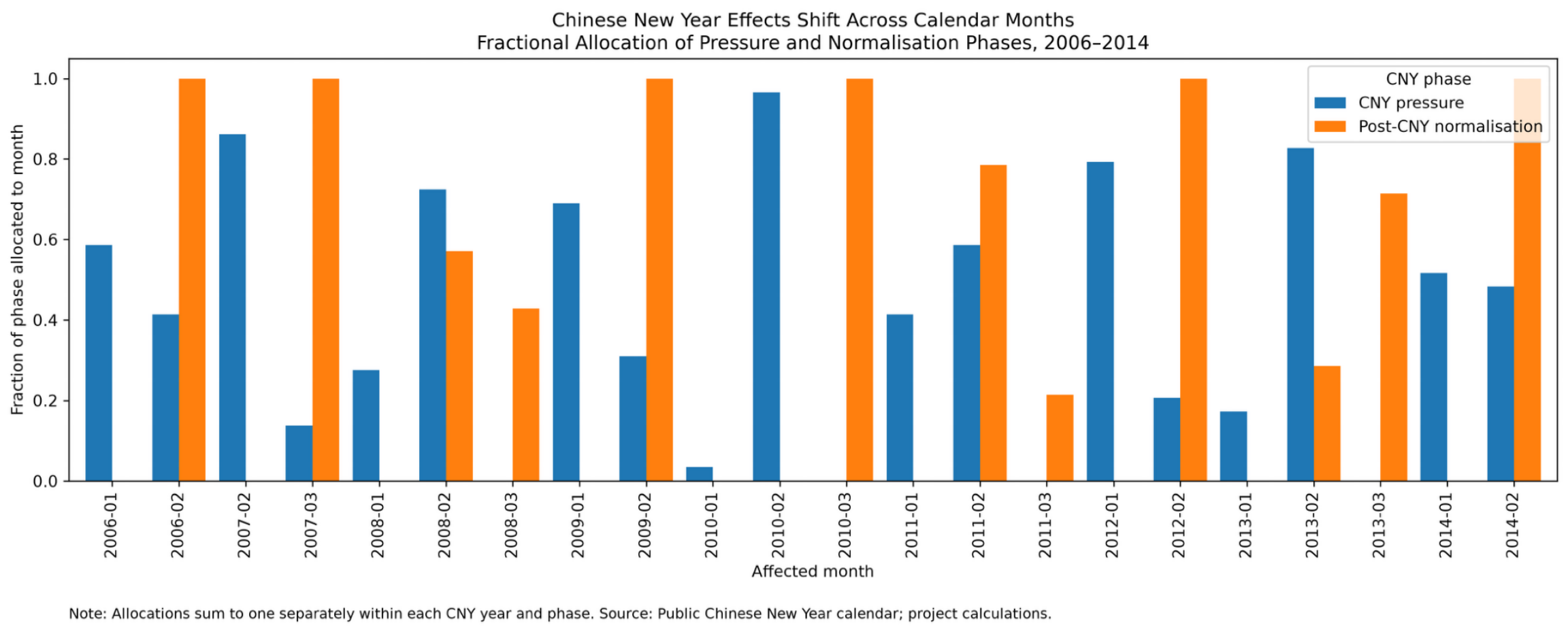
Development-stage choices carried unchanged into final evaluation

Target	Role	Order	Seasonal order	Deterministic terms
Headline	Conventional	(1,0,1)	(1,0,1,12)	Intercept
Headline	CNY-aware MA	(1,0,1)	(0,0,1,12)	Intercept + month + 2 CNY
Headline	CNY-aware AR	(1,0,1)	(1,0,0,12)	Intercept + month + 2 CNY
Core	Primary	(1,0,1)	(1,0,1,12)	Intercept
Core	Robustness	(0,0,2)	(1,0,1,12)	Intercept

No final-evaluation result was used to reopen model selection.

Backup: the CNY phases move across calendar months

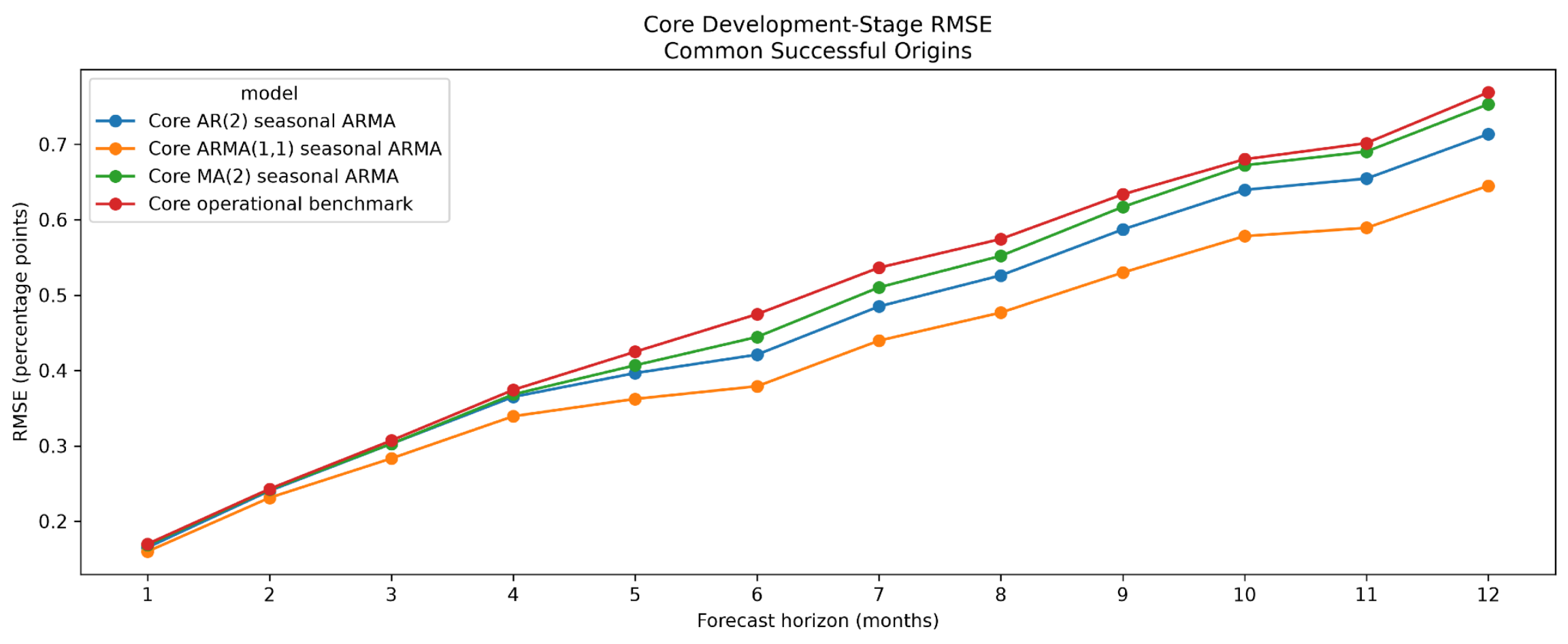
Fractional monthly allocation prevents a fixed-February dummy from misclassifying the holiday effect



Pressure and normalisation are allocated by affected days, then each phase sums to one within year.

Backup: development-stage evidence selected the core primary model

Internal validation precedes and is separate from the final evaluation



The ARMA(1,1) seasonal ARMA led development-stage RMSE, then also led final RMSE from H6 onward.